Project Proposal

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1 What problem are we focusing on?

There exists a lot of nuance to human speech. Most obvious to a listener is one's choice of words, but one's paralinguistic features (e.g. prosody, intonation, use of filler words, etc.) communicate far more than we may expect about their true intent. In particular, **message sincerity** is a nuance that is difficult to capture with only textual data, and recent studies show that a multimodal approach (including text, audio, and other context) supports greater accuracy in deciphering this [CHPR+19, BGP+16].

As humans, we can recognize the level of sincerity in one's speech pretty easily, thus indicating that there are clearly patterns in speech that can help us differentiate the sincere vs. insincere delivery of phrases. The world is very interested in analyzing these patterns in a structured way using machine learning, as evidenced by the growing field of Speech Emotion Recognition (SER) [GM24, BVSR20, GSK+21]. However, there is **very little availability in audio data labelled with perceived message sincerity** [BCHS19]. Additionally, even **unlabelled audio clips of insincerity (i.e. sarcasm) are inherently disproportionately unavailable**, as people tend to be sincere most of the time, so labelling a large amount of data by hand becomes an intractable task. As such, the problem we are focusing on is the lack of available audio data labelled with perceived message sincerity and insincerity.

2 What do we plan to do?

In this research project, we attempt to solve this **data availability** problem by creating a **generative model** for audio clips of different kinds of message sincerity. We plan on looking into VAEs, diffusion models, and latent diffusion models (LDMs), with the objective of reconstructing the original sincere/insincere audio clip.

We will use some existing data from [CHPR+19] ($n = 690$), where each short clip is labelled with a binary "sarcasm" label. We may also supplement this data with our own data, where we would record short clips of us saying the same phrase, in both sincere and insincere (i.e. sarcastic) tones, across multiple takes to model variance within each phrase-tone combination. Data evaluation will inevitably pose a challenge. For both the data we record ourselves and the (potentially unintelligible) data our model generates, we accept the limitation that message sincerity will be subjectively evaluated by us. If time allows, we may try to evaluate our data model against an existing multimodal sarcasm detection dataset (MUStARD) [CHPR+19].

3 What contribution does this make?

If we are successful in creating a useful generative model for sincerity-labelled audio data, it has significant potential for impact in **data augmentation** for future studies in SER research, since the availability of labelled data is currently a bottleneck. Furthermore, if we end up using a lower-dimensional latent space for our generative model, it may be insightful to analyze those latent factors and see if they model any paralinguistic features directly. The latent space may also capture **feature importance** when determining message sincerity, or even message intent in general. In any case, we hope to take one small step towards enabling machines to better understand human speech.

References

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